CUSTOMER CHURN PREDICTION USING EXPLORATORY DATA ANALYSIS AND DEEP LEARNING

PROJECT REPORT

Submitted by

JANARANJAN K(24AD049)

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ABSTRACT

Customer churn is a major challenge in the telecommunications industry, as losing existing customers can affect business growth, revenue, and customer relationships. Identifying the factors that influence customer churn is essential for developing effective customer retention strategies. Exploratory Data Analysis (EDA) provides a systematic approach to understanding customer data by identifying patterns, trends, relationships, and unusual observations through statistical analysis and data visualization. This project, titled **"Customer Churn Prediction Using Exploratory Data Analysis and Deep Learning,"** aims to analyze customer behavior and develop a predictive model to identify customers who are likely to discontinue a service.

The project utilizes the Telco Customer Churn dataset, which contains approximately 7,043 customer records and 21 attributes related to customer demographics, subscribed services, contract types, tenure, monthly charges, total charges, and churn status. Data preprocessing is performed using Python libraries to inspect data types, handle missing values, remove duplicate records, and prepare the dataset for further analysis. Numerical and categorical features are examined to understand their distribution and relationship with the target variable.

Various visualization techniques, including bar charts, histograms, box plots, and correlation heatmaps, are used to identify important patterns in customer behavior. The analysis focuses on churn distribution, contract types, customer tenure, monthly charges, and correlations among numerical variables. These visualizations help identify customer groups that may require further investigation and provide a better understanding of the factors associated with churn. Furthermore, an Artificial Neural Network (ANN) is developed using TensorFlow and Keras to predict whether a customer is likely to leave the service. The dataset is divided into training and testing sets, and categorical variables are encoded while numerical features are scaled. The model is trained using the Adam optimizer and binary cross-entropy loss, with dropout and early stopping techniques used to help reduce overfitting. Model performance is evaluated using accuracy, precision, recall, F1-score, a confusion matrix, and the Receiver Operating Characteristic–Area Under the Curve (ROC-AUC).

The recorded experimental results show an accuracy of approximately 79.06% and a ROC-AUC score of 0.8386. The findings indicate that customer churn patterns are associated with factors such as contract type, tenure, and monthly charges. Overall, this project demonstrates how exploratory data analysis, visualization, and deep learning can be combined to gain meaningful insights from customer data and support data-driven decision-making for customer retention. The results are intended for analytical and educational purposes and may be further improved through model optimization and comparison with other machine learning algorithms.

1. Introduction

Customer churn refers to customers discontinuing a service. Understanding churn is useful for subscription-based businesses because retaining existing customers can be important for long-term growth. Data analysis can reveal patterns in customer behavior, while a supervised deep learning model can estimate the probability that a customer will churn.

2. Objectives

- Explore customer data and identify patterns associated with churn.
- Clean and preprocess the dataset for analysis and modelling.
- Create visualizations to study churn, tenure, contract type, charges, and numerical correlations.
- Develop an Artificial Neural Network to classify customer churn.
- Evaluate model performance using classification metrics and visualizations.
- Discuss how the findings may support customer-retention strategies.

3. Dataset Description

The project uses the Telco Customer Churn dataset, a commonly used customer analytics dataset with 7,043 records and 21 columns. The columns describe customer demographics, subscribed services, contract details, tenure, billing, and churn status. The target variable is Churn, where No represents customers who stayed and Yes represents customers who left. The customerID field is excluded from predictive features because it is an identifier.

Dataset source: https://www.kaggle.com/datasets/blastchar/telco-customer-churn

4. Data Preprocessing

The dataset is loaded with Pandas and inspected for dimensions, column names, data types, missing values, and duplicate rows. The TotalCharges column is converted to numeric because it may be stored as text. Missing values produced by conversion are handled, and duplicate rows are checked before removal. The target variable is mapped to binary labels (No = 0 and Yes = 1). Categorical input columns are encoded using one-hot encoding. The data is split into training and testing sets with an 80:20 ratio and stratified by the target label. StandardScaler is fitted only on the training features and then applied to the test features to reduce data leakage.

5. Exploratory Data Analysis

The following figures are taken from the uploaded notebook outputs. Interpretations below describe visible patterns in these graphs; they do not establish causation.

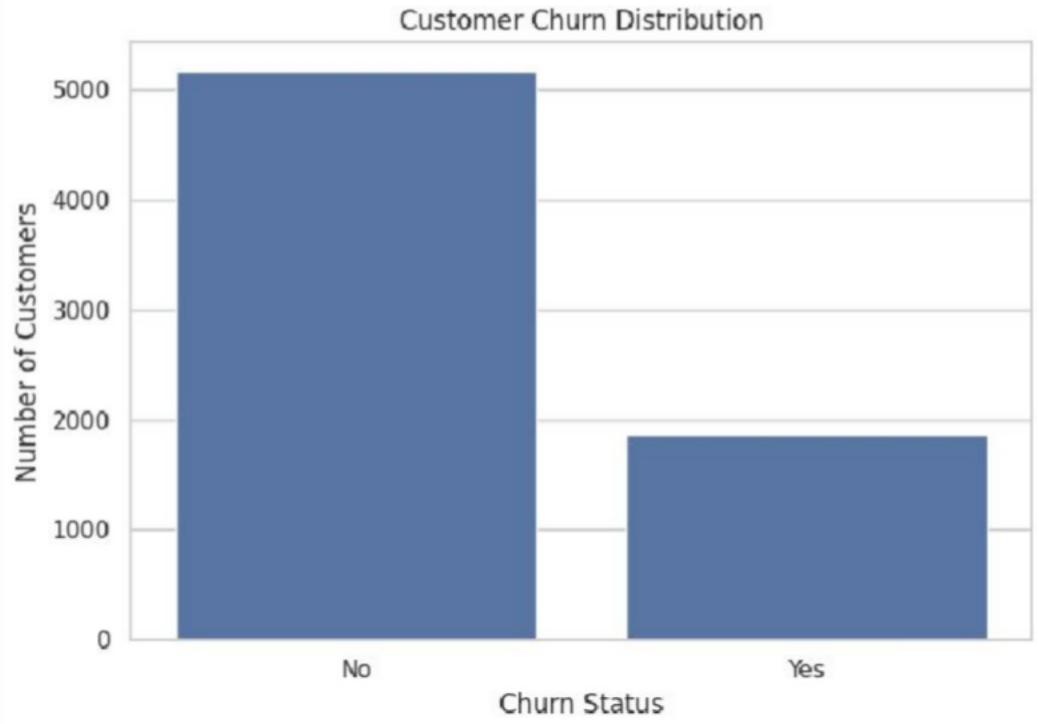


Figure 1. Customer Churn Distribution

Interpretation: The chart shows many more customers in the No Churn category than in the Churn category. This indicates class imbalance, so accuracy alone may not fully describe model performance.

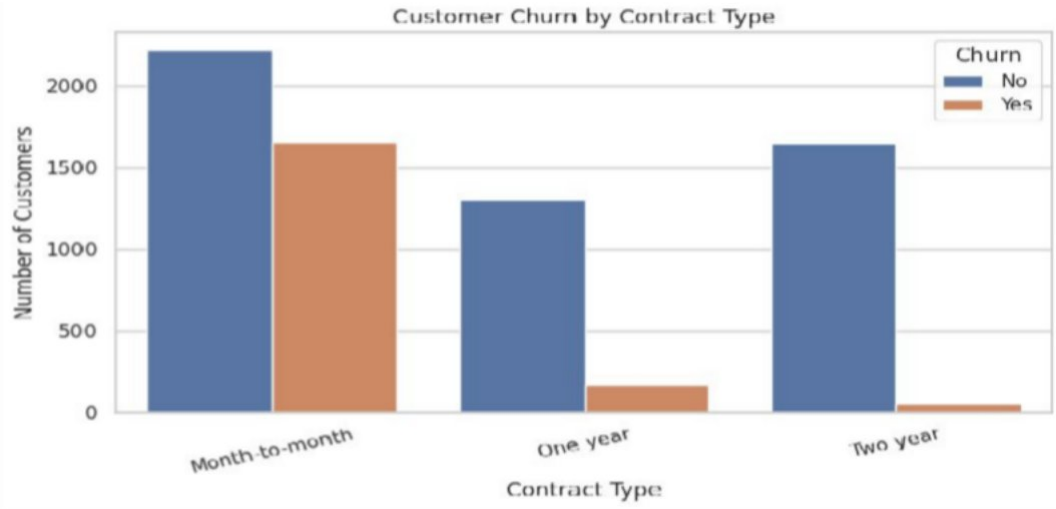


Figure 2. Customer Churn by Contract Type

Interpretation: Month-to-month contracts contain substantially more churn cases than one-year and two-year contracts. The plot displays counts, so comparisons of churn rates should also consider the number of customers in each contract group.

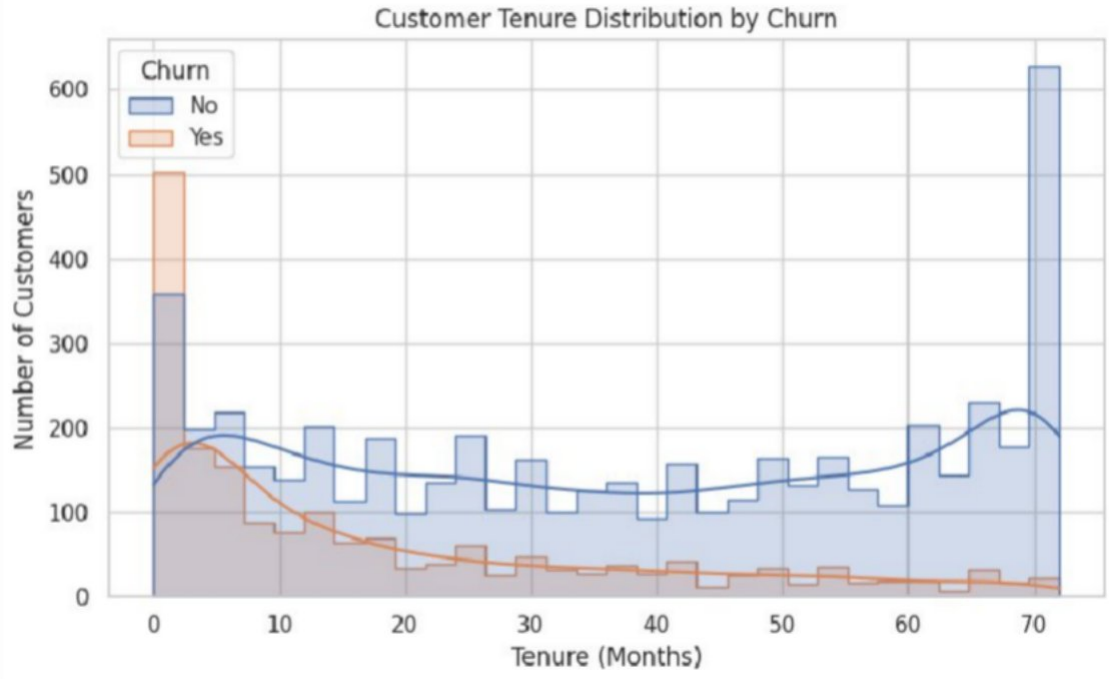


Figure 3. Customer Tenure Distribution by Churn

Interpretation: Churned customers are concentrated more heavily at shorter tenure values, while customers who stayed appear across a broader range of tenure, including longer subscriptions. This suggests that early customer tenure may be an important period for retention.



Figure 4. Monthly Charges by Churn Status

Interpretation: The median monthly charge for customers who churned is higher than the median for customers who stayed in this plot. This is an association in the data and does not prove that higher charges alone cause churn.

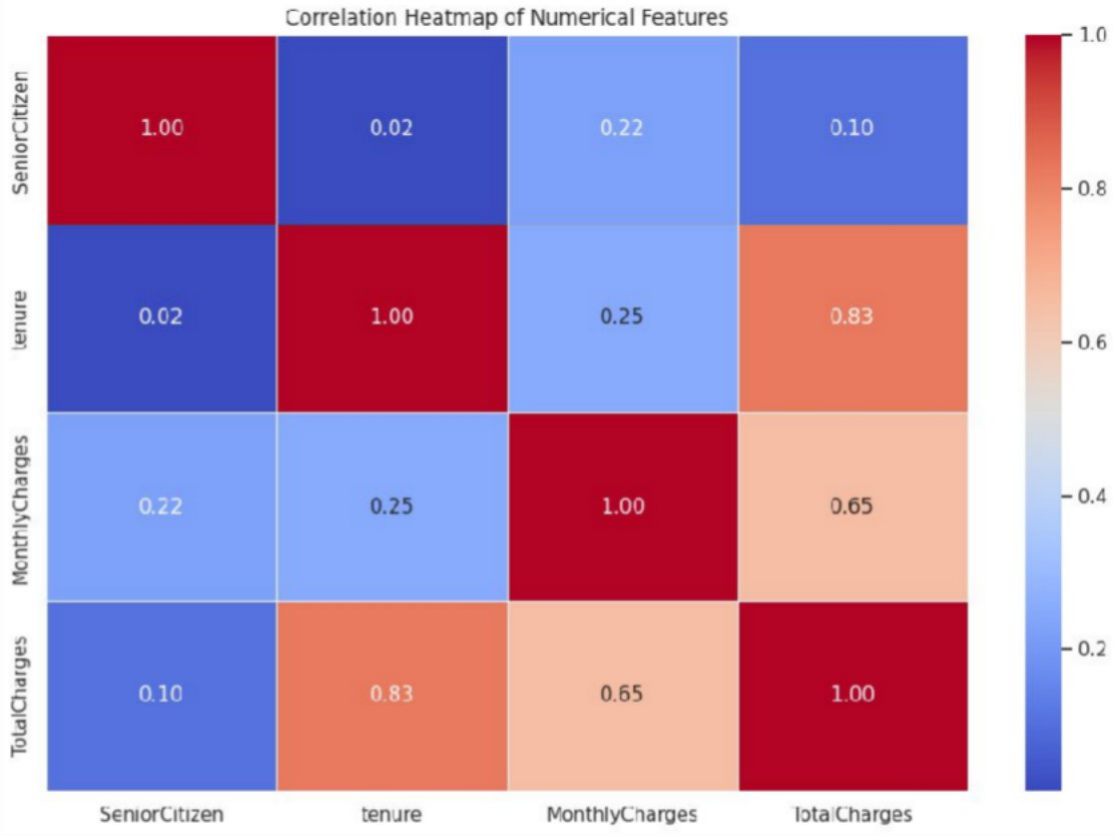


Figure 5. Correlation Heatmap of Numerical Features

Interpretation: Tenure and TotalCharges have a strong positive correlation (0.83), while MonthlyCharges and TotalCharges show a positive correlation (0.65). Total charges tend to accumulate as tenure increases. Correlation values describe linear relationships and do not imply causation.

6. Deep Learning Model

An Artificial Neural Network (ANN) is used for binary classification. The architecture shown in the notebook includes a dense hidden layer with 64 neurons and ReLU activation, a dropout layer, a second dense hidden layer with 32 neurons and ReLU activation, another dropout layer, and a single sigmoid output neuron. The model is compiled with the Adam optimizer and binary cross-entropy loss. Accuracy and AUC are monitored during training, and early stopping is used to restore the weights from the epoch with the best validation loss.

The dataset is split into training and testing subsets. A validation subset is taken from the training data during model fitting. The test set is reserved for final evaluation.

7. Model Evaluation

The notebook's confusion matrix and ROC curve are included below. The supplied notebook images did not include the training/validation accuracy graph, so it is not fabricated here.

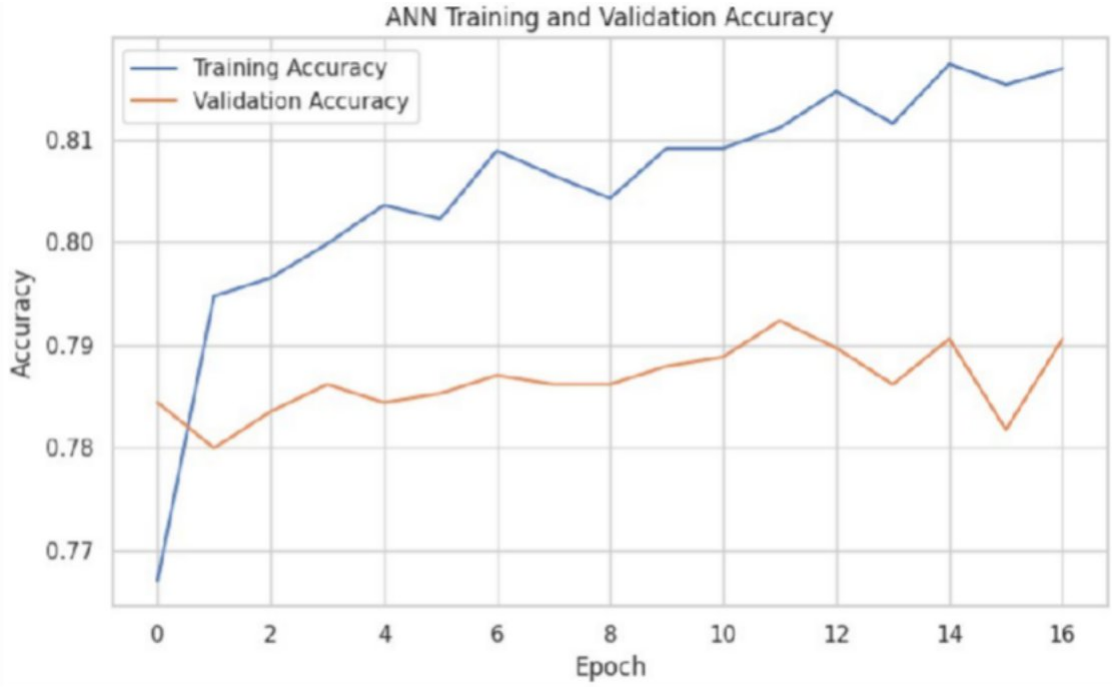


Figure 6. ANN Training and Validation Accuracy

Interpretation:

The graph shows the training and validation accuracy of the Artificial Neural Network (ANN) over multiple epochs. Training accuracy generally increases as the model learns patterns from the training data. Validation accuracy remains relatively stable, with small fluctuations throughout training. The difference between training and validation accuracy suggests that the model learns the training data better than the validation data. Further hyperparameter tuning may improve generalization performance.



Figure7. ANN Training and Validation Loss

Interpretation:

The graph illustrates the training and validation loss of the ANN during model training. Training loss decreases over successive epochs, indicating that the model is learning to minimize prediction errors on the training data. However, validation loss remains relatively constant with minor fluctuations. This difference may indicate overfitting or limited improvement in performance on unseen data. Techniques such as dropout, early stopping, and hyperparameter optimization can help improve the model's generalization ability.

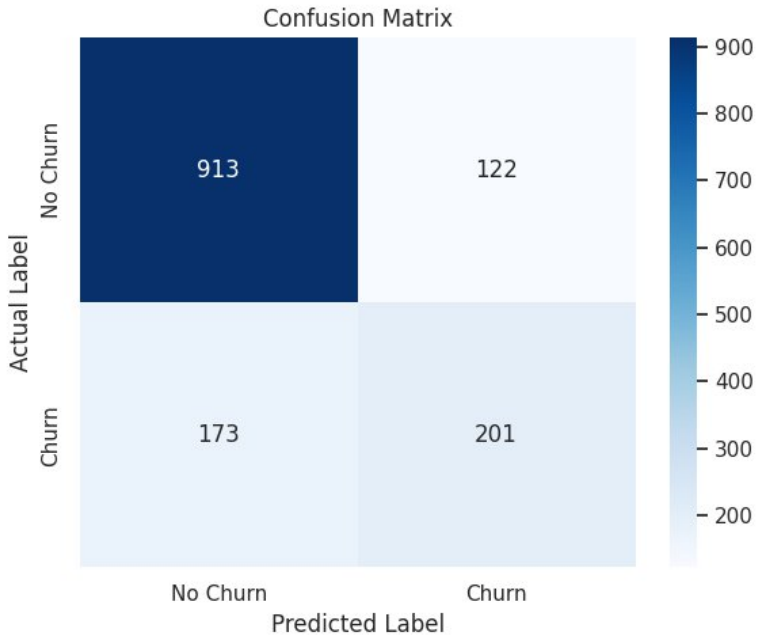


Figure 8. Confusion Matrix

Interpretation:

The confusion matrix evaluates the performance of the Artificial Neural Network (ANN) in classifying customers as churn or non-churn. The model correctly identifies 913 customers who did not churn (True Negatives) and 201 customers who churned (True Positives). However, it incorrectly classifies 122 non-churn customers as churn (False Positives) and fails to identify 173 customers who actually churned (False Negatives).
The results indicate that the model performs better at identifying customers who stay with the service than customers who leave. Improving churn detection and reducing false negatives could help the model support more effective customer retention strategies.

7.1 Evaluation Metrics

The confusion matrix contains 913 true negatives, 122 false positives, 173 false negatives, and 201 true positives. Based on these counts, accuracy is approximately 79.1%, precision for the churn class is approximately 62.2%, recall for the churn class is approximately 53.7%, and F1-score is approximately 57.7%. The ROC curve reports an AUC of approximately 0.839. These values should be cross-checked against the final metrics printed by the before the notebook submission

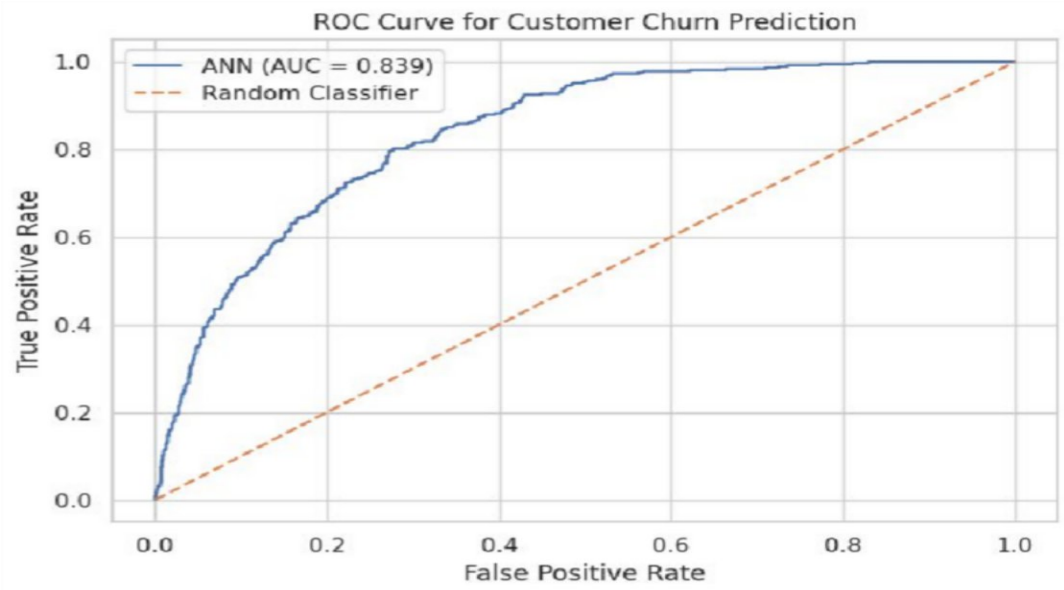


Figure 9. ROC Curve for Customer Churn Prediction

8. Discussion

The visual analysis suggests that churn is associated with contract type, tenure, and monthly charges. Month-to-month customers and customers with shorter tenure appear to be important groups for further investigation. The model's recall indicates that a portion of actual churn cases are missed at the 0.5 probability threshold. For a business application, the prediction threshold could be tuned according to the relative costs of missed churners and unnecessary retention offers.

9. Limitations

- The dataset represents one service-provider context and may not generalize to all businesses.
- The target distribution is imbalanced, so multiple metrics are needed in addition to accuracy.
- The current ANN may benefit from hyperparameter tuning and comparison with baseline models.
- Correlations and observed group differences do not prove causation.
- Results depend on the preprocessing choices, train/test split, random seed, and classification threshold.

10. Future Scope

- Tune the number of neurons, learning rate, batch size, and dropout rate.
- Compare the ANN with logistic regression, random forest, and gradient-boosting models.
- Evaluate class-weighting or resampling methods if appropriate.
- Optimize the classification threshold and examine precision-recall trade-offs.
- Develop a simple dashboard to present churn risk and retention insights.

11. Conclusion

This project demonstrates a workflow for customer churn analysis using EDA and deep learning. The visualizations highlight patterns involving contract type, tenure, and monthly charges. The ANN achieved an AUC of approximately 0.839 in the supplied ROC plot, while the confusion matrix shows that some churn cases remain undetected. Further tuning, threshold selection, and model comparison could improve practical usefulness. The findings should be interpreted as patterns in the dataset rather than proof of causal relationships.

12. References

1. Blastchar. (n.d.). Telco Customer Churn dataset. Kaggle. https://www.kaggle.com/datasets/blastchar/telco-customer-churn
2. Pandas Development Team. (n.d.). Pandas documentation. https://pandas.pydata.org/docs/
3. Scikit-learn developers. (n.d.). Scikit-learn documentation. https://scikit-learn.org/stable/
4. TensorFlow. (n.d.). TensorFlow documentation. https://www.tensorflow.org/
5. Matplotlib Development Team. (n.d.). Matplotlib documentation. https://matplotlib.org/stable/

Appendix A – Notebook

The complete Python code and executed outputs for the Customer Churn Prediction project are provided in the accompanying Google Colab notebook (.ipynb).

The notebook includes dataset loading, data cleaning, exploratory data analysis, visualization, feature preprocessing, ANN model development, model training, and performance evaluation.

GitHub Repository: https://github.com/janaranjan-ai/customer-churn-prediction-deep-learning

The notebook and project files can be accessed through the repository. The notebook is submitted as supporting material for the implementation described in this report.